1. Project Planning & Management

UNSTUCK (AI Automation / Freelance Intelligence Platform)

TRACK & PROGRAM	TARGET MILESTONE	DOCUMENT CODE	ACADEMIC CYCLE
DEPI Technical Track	16 October 2026	DEPI-DOC-01 (M1)	October – December 2026

1. Project Proposal & Executive Overview

Freelancers routinely waste between 1.5 to 3 hours every day manually monitoring fragmented freelance platforms (e.g., Khamsat, Mostaql, Nafezly, Freelancer) sifting through unstructured job descriptions, evaluating skill requirements, and drafting custom bids under tight deadlines.

UNSTUCK is a workflow-driven, intelligent automation ecosystem that completely streamlines this journey. The system automatically harvests job postings across regional and international platforms via resilient scraping pipelines with anti-bot bypass mechanisms, normalizes and persists structured listings in a PostgreSQL datastore, exposes an intuitive web platform with real-time market analytics, and leverages Large Language Models (LLMs) to perform explainable semantic skill-gap matching, automated proposal copywriting, and autonomous smart application submission with flexible manual oversight.

1.1 Project Objectives

- **Automated Ingestion Pipeline:** Eliminate repetitive manual searching through scheduled n8n workflows and multiplatform scrapers (Khamsat, Mostaql, Nafezly, Freelancer).
- **Structured Relational Datastore:** Enforce unified schemas with currency normalization, B-tree indexing, and zeroduplicate guarantees using content hashes.
- **Interactive Web Platform:** Provide a modern React / Tailwind UI for searching, multi-criteria filtering, freelancer profile management, and market analytics.
- **AI Semantic Skill Matching:** Benchmark job requirements against freelancer profiles using LLMs to output transparent match scores, matched competencies, and missing skills.
- **Automated Proposal & Auto-Apply Engine:** Generate tailored client proposals and execute autonomous smart bidding via a coordinated multi-agent AI architecture with human oversight options.

1.2 Project Scope (100% In-Scope Delivery)

> **Official Scope Status: 100% In-Scope (Out of Scope: None / 0%)**
> All data acquisition, anti-bot handling, full-stack platform features, AI matching, proposal generation, and auto-bidding capabilities are fully integrated into the approved graduation scope.

Scope Category	Integrated Feature Set	Status
Data Acquisition & Scraping	Multi-platform collection across Khamsat, Mostaql, Nafezly, and Freelancer with proxy rotation and anti-bot bypass.	100% In-Scope
Data Engineering & Storage	PostgreSQL schema, hash-based deduplication (MD5/SHA256), currency normalization, and Redis query caching.	100% In-Scope
Web Application & APIs	Full-stack responsive interface (React / Tailwind / FastAPI) with cataloging, live filtering, and market analytics dashboard.	100% In-Scope
AI Intelligence & Matching	LLM-driven semantic skill extraction, explainable skill-gap scoring, and candidate suitability benchmarking.	100% In-Scope
Autonomous Application & Bidding	Multi-Agent AI architecture orchestrating automated proposal copywriting, optimal bid pricing, and autonomous submission.	100% In-Scope

2. Official DEPI Milestone Timeline (October – December 2026)

In strict compliance with the **Digital Egypt Pioneers Initiative (DEPI) Official Guidelines** (distinguished from the compressed DEPI schedule), the graduation project follows the full 6-milestone roadmap extending to December 2026:

DEPI Milestone / Deliverable	Official DEPI Target Date	Current Status

1. Project Planning & Management	**16 Oct 2026**	**Completed (M1)**
2. Literature Review	**16 Oct 2026**	**Completed (M2)**
3. Requirements Gathering	**16 Oct 2026**	**Completed (M3)**
4. System Analysis & Design	**06 Nov 2026**	**In Progress**
5. Implementation (Source Code & Execution)	**30 Nov 2026**	**Scheduled**
6. Final Presentation & Testing & Reports	**04 Dec 2026**	**Scheduled**

2.1 Four Technical Implementation Phases

Phase	Focus Area	Key Deliverables & Activities	Timeline
Phase 1	Project Study & Database Setup	System architecture, Multi-Agent spec, PostgreSQL ERD, schema scripts, and QA test plan.	12 – 26 Oct
Phase 2	Multi-Platform Scraper & Ingestion	n8n harvesting workflows, Playwright or beautifulsoup anti-bot scraper modules, currency sanitizer, deduplication.	26 Oct – 09 Nov
Phase 3	Frontend, Backend & Analytics	FastAPI REST endpoints, React / Tailwind catalog, filter sidebars, market analytics charts.	09 – 20 Nov
Phase 4	AI Matching, Auto-Apply & Delivery	LLM skill matching, proposal copywriter, auto-bidding worker, QA regression suite, final presentation.	20 – 30 Nov

3. Team Roles & Responsibilities (RACI Matrix)

The project is executed by a dedicated four-member engineering team with clearly distributed responsibilities:

Team Member & Title	Specialized Role	Core Engineering Responsibilities
Ahmed Mohamed Saad Elshahat	Team Leader & AI/Automation Engineer	Overall project leadership, n8n orchestration design, Multi-Agent architecture, LLM prompt engineering, proposal copywriter, autobidding coordination.
Aya El-sayed Abd El-Fadel	Data Engineer & Backend Developer	PostgreSQL schema design (ERD), data normalization, hash deduplication, FastAPI REST endpoints, submission worker endpoints, and Redis caching.
Samira Ali Abdel Gaffar	Frontend Developer & UI/UX Specialist	React / Tailwind UI development, job catalog dashboard, filter and search components, freelancer profile manager, proposal review modal, analytics charts.
Iman Ahmed Elmetwali Mohammed	Data Collection Specialist & QA/Testing	Target platform scrapers (Khamsat, Mostaql, Nafezly, Freelancer), anti-bot/proxy rotation, test case design, integration testing, QA verification logs.

3.1 Project RACI Matrix

Deliverable / Major Activity	Ahmed M. Saad (AI/Lead)	Aya El-sayed (Data/Backend)	Samira Ali (Frontend)	Iman Ahmed (Scraping/QA)
Project Planning & Management Specification	**Accountable / Resp.**	Consulted	Consulted	Consulted
Database Schema Design (ERD) & Normalization	Accountable	**Responsible**	Consulted	Informed
Multi-Platform Scrapers & Anti-Bot Bypass	Consulted	Informed	Informed	**Responsible**

n8n Ingestion Workflows & Orchestration	**Responsible**	Consulted	Informed	Informed
Backend REST API & Analytics Endpoints	Accountable	**Responsible**	Consulted	Informed
Web Platform UI & Analytics Dashboard	Accountable	Consulted	**Responsible**	Informed
AI Semantic Matching & Proposal Generation	**Responsible**	Consulted	Consulted	Informed

Deliverable / Major Activity	Ahmed M. Saad (AI/Lead)	Aya El-sayed (Data/Backend)	Samira Ali (Frontend)	Iman Ahmed (Scraping/QA)
Autonomous Submission Engine (Auto-Apply)	**Responsible**	**Responsible**	Consulted	Informed
Quality Assurance & End-to-End Verification	Accountable	Consulted	Consulted	**Responsible**

4. Risk Assessment & Mitigation Plan

ID	Identified Risk	Severity	Likelihood	Engineered Mitigation Strategy
RSK-01	Target platform DOM layout changes or rate-limiting.	High	Medium	Employ modular scraper adapters, proxy pool rotation, Playwright stealth mode, and scheduled polling intervals.
RSK-02	Duplicate postings polluting database quality.	Medium	High	Enforce unique constraints on MD5 content hashes (URL + Title) in PostgreSQL with idempotent upserts.
RSK-03	Excessive LLM token costs during high-volume matching.	Medium	Medium	Pre-filter jobs with keyword matching before AI inference; cache match results in Redis; utilize efficient models.
RSK-04	Autonomous application submission failure or captcha locks.	High	Low	Incorporate human-in-the-loop review checkpoints, strict retry thresholds, and verification confirmation logs.
RSK-05	Schedule slippage across 6 milestones.	High	Low	Adhere strictly to official DEPI target dates (Oct 16, Nov 6, Nov 30, Dec 4) with weekly milestone reviews.

5. Key Performance Indicators (KPIs) & Success Criteria

KPI Category	Target Metric	Success Criterion	Verification Method
Data Quality	Duplicate Insertion Rate = 0%	100% deduplication of scraped listings	PostgreSQL UNIQUE hash constraint checks.
System Latency	REST API Response < 250ms	Sub-second filtering & catalog browsing	Network profiling & API stress testing.
AI Matching Precision	Structured JSON Schema = 100%	Valid skill-gap scoring with explanations	Pydantic output schema validation tests.
Workflow Reliability	Pipeline Uptime ≥ 98%	Scheduled n8n harvesting without data loss	Automated execution logging in PostgreSQL.